Problem Set 2

Due January 25, 2024 at 11:59 PM Pacific

Turn in your problem set by Gradescope. Include any code you write in your solutions as well.

1. Using the definition of expectation, we will verify key properties for an n-dimensional Gaussian random vector X with distribution:

$$p(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-1/2(x-\mu)^T \Sigma^{-1}(x-\mu)}.$$

In past classes, you may have proved these properties for a 1D Gaussian random variable, but we must understand these properties in this course's multidimensional setting. If you have not seen the proof of 1D Gaussian properties, we recommend consulting the following website: `https://www.statlect.com/probability-distributions/normal-distribution`

(a) First, prove that $\mathbb{E}[X] = \mu$.

Hint 1: Consider transforming to the multivariate standard Gaussian distribution discussed in lecture. Recall that for a multidimensional, linear change of variables, if $u = Ax + b$, then $du = \det(A)\, dx$. The $\det(A) = \frac{du}{dx}$ accounts for the change in n-dimensional volume from the change of variables. If you are having trouble, consider first showing that the standard multidimensional Gaussian random vector has a mean equal to the zero vector.

Hint 2: You may find yourself doing a bit of linear algebra. This is completely acceptable. You may alternatively find it helpful to consider properties of integrals of odd or even functions.

(b) Second, prove that $\mathbb{E}[(X-\mu)(X-\mu)^T] = \Sigma$

Hint 1: You may want to use the same change of variable as above. Again, if you are having trouble, consider first showing that the standard multidimensional Gaussian random vector has a covariance equal to the identity matrix.

Hint 2: To simplify your expression, you will want to integrate by parts or invoke properties of the 1D standard Gaussian (proven in the website above).

2. Show the following:

(a) The product of two Gaussian PDFs is an unnormalized Gaussian, $\mathcal{N}(\mu_1, \Sigma_1)\mathcal{N}(\mu_2, \Sigma_2) = \eta\mathcal{N}(\mu, \Sigma)$, for some constant η. Give the expressions for μ and Σ in terms of μ_1, μ_2, Σ_1, and Σ_2.

Hint: Note that $X \sim \mathcal{N}(\mu, \Sigma)$ means that $p(x) = \mathcal{N}(\mu, \Sigma)$ is as given in the previous problem.

(b) $\mathbb{E}[(X-\mu)(X-\mu)^T] = \mathbb{E}[XX^T] - \mu\mu^T$, where $\mu := \mathbb{E}[X]$. Hence $\mathrm{Cov}(X)$ can be written in either of these two forms.

(c) If two RVs X and Y are independent, they are uncorrelated, $\mathrm{Cov}(X, Y) = 0$. That is, independence is *stronger* than uncorrelated. (Note: The converse is not true in general, but it is true for jointly Gaussian RVs. But you don't have to show this.)

Hint: You may want to use part (b) above.

3. For a Gaussian random vector $X \sim \mathcal{N}(\mu, \Sigma)$, we call the region $\mathcal{B}(\epsilon) = \{x \mid p(x) \geq \epsilon\}$ the error ellipse of $\mathcal{N}(\mu, \Sigma)$.

(a) Show that $\mathcal{B}(\epsilon)$ defines an ellipse in $\mathbb{R}^n$ by expressing it in a quadratic form in x with a positive definite matrix.

Hint: A quadratic form in x is $(x+v)^T M(x+v)$ for matrix M and vector v.

(b) For the 2D case, show that

$$\epsilon = \frac{1-P}{2\pi|\Sigma|^{1/2}}$$

gives the error ellipse with the property $Pr\big(X \in \mathcal{B}(\epsilon)\big) = P$ for a given probability $0 < P < 1$.

Hint: As in problem 1, use the parameterization $w = \Sigma^{-1/2}(x-\mu)$ that we saw in class to convert the Gaussian to a standard form, then integrate.

4. Write a function in `Python` to calculate a 2D error ellipse you found above with $P = 0.95$, given input parameters (μ, Σ). Plot 2D error ellipses overlying 1000 samples drawn from the distribution for three different 2D Gaussians of your choice. Validate your ellipse by calculating the proportion of the 1000 points that lie inside the ellipse (it should be close to P). Attach your plots.

Hint: The Python functions `numpy.random.randn` and `scipy.linalg.sqrtm` will come in handy.

Hint: You already have the equation for calculating if the point is internal or external to the plotted ellipse from question 3a.

1. Using the definition of expectation, we will verify key properties for an n-dimensional Gaussian random vector X with distribution:

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Hint 2: You may find yourself doing a bit of linear algebra. This is completely acceptable. You may alternatively find it helpful to consider properties of integrals of odd or even functions.

(b) Second, prove that $\mathbb{E}[(X-\mu)(X-\mu)^T] = \Sigma$

Hint 1: You may want to use the same change of variable as above. Again, if you are having trouble, consider first showing that the standard multidimensional Gaussian random vector has a covariance equal to the identity matrix.

Hint 2: To simplify your expression, you will want to integrate by parts or invoke properties of the 1D standard Gaussian (proven in the website above).

$$E[x] = \int_x x\, p(x)\, dx$$

$$= \int_x x \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} dx$$

$$w = \Sigma^{-\frac{1}{2}}(x-\mu) \qquad x = \Sigma^{\frac{1}{2}} w + \mu$$

$$w^T = (x-\mu)^T \Sigma^{-\frac{1}{2}T} = (x-\mu)^T \Sigma^{-\frac{1}{2}} \quad \text{(but symmetrical)} \qquad dx = |\Sigma^{\frac{1}{2}}|\, dw$$

$$\therefore \quad E[x] = \int_w \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} (\Sigma^{\frac{1}{2}} w + \mu) e^{-\frac{1}{2} w^T w} |\Sigma^{\frac{1}{2}}|\, dw$$

$|\Sigma^{\frac{1}{2}}| = |\Sigma|^{\frac{1}{2}}$ for symmetric positive definite

$$= \int_w \frac{1}{\sqrt{(2\pi)^n}} w\, e^{-\frac{1}{2} w^T w} dw + \mu \int_w \frac{1}{\sqrt{(2\pi)^n}} e^{-\frac{1}{2} w^T w} dw$$

(odd function)

$$= \mu \int \prod_{k=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} w_k^2} dw$$

$$\frac{1}{\sqrt{(2\pi)^n}} e^{-\frac{1}{2}\|w\|^2} = \prod_{k=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} w_k^2}$$

$$= \mu \left[\frac{1}{\sqrt{2\pi}} \sqrt{2\pi}\right]^n$$

$$\boxed{E[x] = \mu}$$

LOTUS

b) $$E[(x-\mu)(x-\mu)^T] = \int_x (x-\mu)(x-\mu)^T p(x)\, dx$$

$$= \frac{1}{(2\pi)^{n/2} |\Sigma|^{\frac{1}{2}}} \int (x-\mu)(x-\mu)^T e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} dx$$

$x = \Sigma^{\frac{1}{2}} w + \mu$

$dx = |\Sigma^{\frac{1}{2}}|\, dw$

$$= \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \int (\Sigma^{1/2} w)(\Sigma^{1/2} w)^T e^{-\frac{1}{2} w^T w} |\Sigma^{1/2}| dw$$

$|\Sigma^{1/2}| = |\Sigma|^{1/2}$ pos. semi-def.

$(\Sigma^{1/2})^T = \Sigma^{1/2}$ symmetric

$$= \Sigma^{1/2} \left[\int \frac{1}{(2\pi)^{n/2}} w w^T e^{-\frac{1}{2} w w^T} dw \right] \Sigma^{1/2}$$

$ww^T = w_i w_j$ element-wise → split into sum over $i, j \in 1:n$

$$= \int_{w_j} \prod_{i=j}^{n} \int_{w_i} \prod_{i=1}^{n} \frac{1}{(2\pi)^{n/2}} w_i w_j e^{-\frac{1}{2}\|w\|^2} dw_i dw_j$$

for $i \neq j$

$$\int_{w_j} \int_{w_i} \frac{1}{(2\pi)^{n/2}} w_i w_j e^{-\frac{1}{2}\|w\|^2} dw_i dw_j = 0$$

odd function on w_i

for $i = j$

$$\int_{w_i} \frac{1}{(2\pi)^{n/2}} w_i^2 e^{-\frac{1}{2}\|w\|^2} dw_i = \left[\int_{w_i} \frac{1}{\sqrt{2\pi}} w_i^2 e^{-\frac{1}{2} w_i^2} dw_i \right] \prod_{k \neq i}^{n} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} w_k^2} dw_k$$

split these

even function

even function

$$\frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}\|w\|^2} = \prod_{k=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} w_k^2}$$

$$= \frac{1}{\sqrt{2\pi}} (\sqrt{2\pi}) \left\{ \frac{1}{(2\pi)^{n-1}} \right\} = 1$$

$$\therefore \quad E[(x-\mu)(x-\mu)^T] = \Sigma^{1/2} \cdot \Sigma^{1/2}$$

$$\boxed{E[(x-\mu)(x-\mu)^T] = \Sigma}$$

2. Show the following:

(a) The product of two Gaussian PDFs is an unnormalized Gaussian, $\mathcal{N}(\mu_1, \Sigma_1)\mathcal{N}(\mu_2, \Sigma_2) = \eta\mathcal{N}(\mu, \Sigma)$, for some constant η. Give the expressions for μ and Σ in terms of μ_1, μ_2, Σ_1, and Σ_2.

Hint: Note that $X \sim \mathcal{N}(\mu, \Sigma)$ means that $p(x) = \mathcal{N}(\mu, \Sigma)$ is as given in the previous problem.

(b) $\mathbb{E}[(X-\mu)(X-\mu)^T] = \mathbb{E}[XX^T] - \mu\mu^T$, where $\mu := \mathbb{E}[X]$. Hence $\mathrm{Cov}(X)$ can be written in either of these two forms.

(c) If two RVs X and Y are independent, they are uncorrelated, $\mathrm{Cov}(X,Y) = 0$. That is, independence is *stronger* than uncorrelated. (Note: The converse is not true in general, but it is true for jointly Gaussian RVs. But you don't have to show this.)

Hint: You may want to use part (b) above.

a) if $p(x) = \mathcal{N}(\mu, \Sigma)$

$$\mathcal{N}(\mu_1, \Sigma_1)\mathcal{N}(\mu_2, \Sigma_2) = p_1(x)\,p_2(x)$$

$$= \frac{1}{\sqrt{(2\pi)^n |\Sigma_1|}} e^{-\frac{1}{2}(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1)} \frac{1}{\sqrt{(2\pi)^n |\Sigma_2|}} e^{-\frac{1}{2}(x-\mu_2)^T \Sigma_2^{-1}(x-\mu_2)}$$

can be put into η

$$= \eta\, e^{-\frac{1}{2}(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1)} e^{-\frac{1}{2}(x-\mu_2)^T \Sigma_2^{-1}(x-\mu_2)} = e^{-\frac{1}{2}(\text{sum of both})}$$

$$(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1) + (x-\mu_2)^T \Sigma_2^{-1}(x-\mu_2)$$

$$= x^T\Sigma_1^{-1}x - \mu_1^T\Sigma_1^{-1}x - x^T\Sigma_1^{-1}\mu_1 + \mu_1^T\Sigma_1^{-1}\mu_1 + x^T\Sigma_2^{-1}x - \mu_2^T\Sigma_2^{-1}x - x^T\Sigma_2^{-1}\mu_2 + \mu_2^T\Sigma_2^{-1}\mu_2$$

$$= x^T(\Sigma_1^{-1} + \Sigma_2^{-1})x - (\mu_1^T\Sigma_1^{-1} + \mu_2^T\Sigma_2^{-1})x - x^T(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2) + (\mu_1^T\Sigma_1^{-1}\mu_1 + \mu_2^T\Sigma_2^{-1}\mu_2)$$

want:

$$(x-\mu_3)^T\Sigma_3^{-1}(x-\mu_3) = x^T\Sigma_3^{-1}x - \mu_3\Sigma_3^{-1}x - x^T\Sigma_3^{-1}\mu_3 + \mu_3^T\Sigma_3^{-1}\mu_3$$

$\rightarrow \Sigma_1^{-1} + \Sigma_2^{-1} = \Sigma_3^{-1}$ $\quad \frac{1}{\Sigma_1} + \frac{1}{\Sigma_2} = \frac{1}{\Sigma_3}$

$$\boxed{\Sigma = \frac{\Sigma_1\Sigma_2}{\Sigma_1 + \Sigma_2}}$$

$\rightarrow \Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2 = \Sigma_3^{-1}\mu_3$ $\quad \frac{\mu_1}{\Sigma_1} + \frac{\mu_2}{\Sigma_2} = \frac{\mu}{\Sigma} = \frac{\Sigma_1+\Sigma_2}{\Sigma_1\Sigma_2}\mu$

$$\frac{\mu_1\Sigma_2 + \mu_2\Sigma_1}{\Sigma_1\Sigma_2} = \frac{(\Sigma_1+\Sigma_2)\mu}{\Sigma_1\Sigma_2}$$

$$\boxed{\mu = \frac{\mu_1\Sigma_2 + \mu_2\Sigma_1}{\Sigma_1\Sigma_2}}$$

b) $E[(X-\mu)(X-\mu)^T] = \int_x (x-\mu)(x-\mu)^T p(x)\,dx$

$= \int (xx^T - \mu x^T - x\mu^T + \mu\mu^T) p(x)\,dx$

$= \int xx^T p(x)\,dx - \mu \int x^T p(x)\,dx - \int x\, p(x)\,dx\, \mu^T + \mu\mu^T \int p(x)\,dx$

$= E[xx^T] - \mu E[x^T] - E[x]\mu^T + \mu\mu^T$

($E[x^T] = \mu^T$, $E[x] = \mu$)

$= E[xx^T] - \mu\mu^T - \mu\mu^T + \mu\mu^T = \boxed{E[xx^T] - \mu\mu^T}$

c) $Cov(X,Y) = E[(X-\mu_x)(Y-\mu_y)^T] = \int_x \int_y (X-\mu_x)(Y-\mu_y)^T p(x,y)\,dy\,dx$

but for independent, $p(X,Y) = p(X)p(Y)$

$cov(x,y) = \int_x \int_y (X-\mu_x)(Y-\mu_y)^T p(x)p(y)\,dy\,dx$

can put $p(x)$ outside

inner $\int_y (XY - \mu_x Y - X\mu_y + \mu_x\mu_y) p(y)\,dy$

$= \int_y XY p(y)\,dy - \mu_x \int_y Y p(y)\,dy - \int_y X p(y)\,dy\,\mu_y + \mu_x\mu_y \int_y p(y)\,dy$

($\int_y p(y)\,dy = 1$, $\int_y X p(y)\,dy = X$)

$= X E[Y] - \mu_x E[Y] - X\mu_y + \mu_x\mu_y$

($E[Y] = \mu_y$, $E[Y] = \mu_y$)

$= X\mu_y - X\mu_y = \boxed{0}$

3. For a Gaussian random vector $X \sim \mathcal{N}(\mu, \Sigma)$, we call the region $\mathcal{B}(\epsilon) = \{x \mid p(x) \geq \epsilon\}$ the error ellipse of $\mathcal{N}(\mu, \Sigma)$.

(a) Show that $\mathcal{B}(\epsilon)$ defines an ellipse in $\mathbb{R}^n$ by expressing it in a quadratic form in x with a positive definite matrix.

Hint: A quadratic form in x is $(x+v)^T M(x+v)$ for matrix M and vector v.

(b) For the 2D case, show that

$$\epsilon = \frac{1-P}{2\pi|\Sigma|^{1/2}}$$

gives the error ellipse with the property $Pr(X \in \mathcal{B}(\epsilon)) = P$ for a given probability $0 < P < 1$.

Hint: As in problem 1, use the parameterization $w = \Sigma^{-1/2}(x-\mu)$ that we saw in class to convert the Gaussian to a standard form, then integrate.

a) $p(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$

just replacing as constant $= \eta\, e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$

η is always positive

$p(x) \geq \epsilon \rightarrow \eta\, e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} \geq \epsilon \quad \therefore \quad -\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu) \geq \ln\left(\frac{\epsilon}{\eta}\right)$

so $\boxed{(x-\mu)^T \Sigma^{-1}(x-\mu) \leq -2\ln\left(\frac{\epsilon}{\eta}\right)}$ quadratic form. $\leq$ because it is inside the ellipse

b) $P = \int_{x \in \mathcal{B}} p(x)\,dx = \int_{x \in \mathcal{B}} \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}\,dx$

$w = \Sigma^{-\frac{1}{2}}(x-\mu)$

$x = \Sigma^{\frac{1}{2}} w + \mu$

$w^T = (x-\mu)^T \Sigma^{-\frac{1}{2}T} = (x-\mu)^T \Sigma^{-\frac{1}{2}}$ (but symmetrical)

$dx = |\Sigma^{\frac{1}{2}}|\,dw$

$\int_w \frac{1}{2\pi |\Sigma|^{1/2}} e^{-\frac{1}{2} w^T w} |\Sigma^{\frac{1}{2}}|\,dw = \frac{1}{2\pi} \int_{w^2 \leq R^2} e^{-\frac{1}{2} w^T w}\,dw$ ($|\Sigma^{1/2}| = |\Sigma|^{1/2}$)

odd function

polar $dw = r\,dr\,d\theta$

$P = \frac{1}{2\pi} \int_0^{2\pi} d\theta \int_0^R r e^{-\frac{1}{2}r^2}\,dr = \int_0^R e^{-\frac{1}{2}r^2} r\,dr = \left(\frac{1}{2}\right) \int_0^{R^2} e^{-\frac{u}{2}}\,du = \left(\frac{1}{2}\right)\left(-2e^{-\frac{u}{2}}\right)\Big|_0^{R^2}$ ($u = r^2$, $du = 2r\,dr$; 2π)

$P = 1 - e^{-\frac{R^2}{2}}$

$(x-\mu)^T \Sigma^{-1}(x-\mu) \leq -2\ln\left(\frac{\epsilon}{\eta}\right)$

$\rightarrow w^T w \leq -2\ln\left(\frac{\epsilon}{\eta}\right) = R^2$

$P = 1 - e^{-\frac{1}{2}\left(-2\ln\frac{\epsilon}{\eta}\right)} = 1 - \frac{\epsilon}{\eta}$

$\epsilon = \eta(1-P) = \frac{1}{2\pi |\Sigma|^{\frac{1}{2}}}(1-P)$

$\boxed{\epsilon = \frac{1-P}{2\pi |\Sigma|^{1/2}}}$

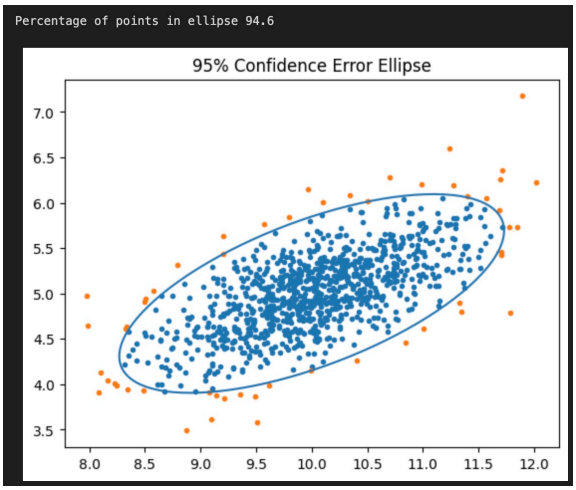
4. Write a function in `Python` to calculate a 2D error ellipse you found above with $P = 0.95$, given input parameters (μ, Σ). Plot 2D error ellipses overlying 1000 samples drawn from the distribution for three different 2D Gaussians of your choice. Validate your ellipse by calculating the proportion of the 1000 points that lie inside the ellipse (it should be close to P). Attach your plots.

Hint: The Python functions `numpy.random.randn` *and* `scipy.linalg.sqrtm` *will come in handy.*

Hint: You already have the equation for calculating if the point is internal or external to the plotted ellipse from question 3a.

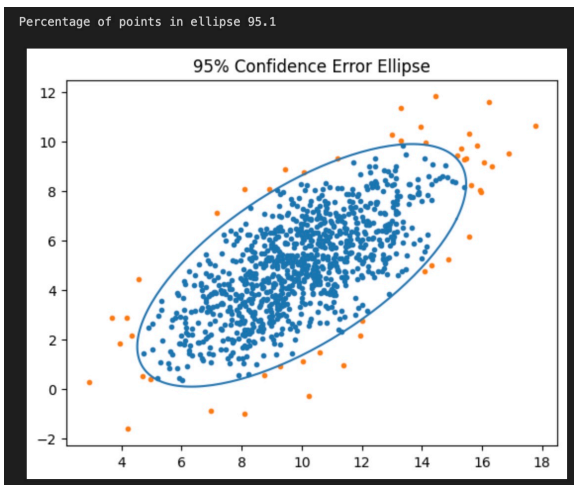
$$\mu = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 0.5 & 0.2 \\ 0.2 & 0.2 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 5 & 3 \\ 3 & 4 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1.4 & 3 \\ 3 & 8 \end{bmatrix}$$

